PROCEDURE

1. Login and Folder Creation

1. Log in to the Cadence system using the assigned student account credentials.

2. Open the **Home** folder, followed by the **Cadwork** folder and then the **Digital** folder.

3. Create a new folder using the student USN as its name.

4. Inside the USN folder, create a folder named **sim**.

5. Inside the **sim** folder, create three subfolders: **rtl** (Verilog design files), **tb** (testbench files), and **Work** (simulation outputs and waveform files).

2. Open the Terminal and Configure the Environment

1. Return to the **Digital** folder and open the terminal.

```
ls -a
csh
source .cshrc
which ncverilog
cd USN/
ls
cd sim/
ls
```

2. The `which ncverilog` command checks whether the NC-Verilog simulator is available in the configured environment.

3. Create the RTL Design Files

1. Navigate to the **rtl** folder:

```
cd rtl/
```

2. Create the Verilog file for the 4-bit Magnitude Comparator:

```
gedit comparator_4bit.v
```

3. Enter the comparator Verilog design code, save the file and close the editor.

4. Create the Verilog file for the 4-bit Ring Counter:

```
gedit ring_counter.v
```

5. Enter the ring counter Verilog design code, save the file and close the editor.

6. Verify that the files have been created:

```
ls -a
```

4. Create the Testbench Files

1. Navigate to the **tb** folder:

```
cd ../tb/
```

2. Create the comparator testbench:

```
gedit comparator_4bit_tb.v
```

3. Enter the comparator testbench code, save the file and close the editor.

4. Create the ring counter testbench:

```
gedit ring_counter_tb.v
```

5. Enter the ring counter testbench code, save the file and close the editor.

6. Verify that both testbench files are present:

```
ls -a
```

5. Compile and Simulate the Designs

1. Navigate to the **Work** folder:

```
cd ../Work/
```

2. Compile and simulate the 4-bit Magnitude Comparator using:

```
ncverilog +access+r ../rtl/comparator_4bit.v ../tb/comparator_4bit_tb.v
```

3. Observe the simulation output in the terminal. The comparator should display the greater-than, equal-to and less-than results for the applied input combinations.

4. Compile and simulate the 4-bit Ring Counter using:

```
ncverilog +access+r ../rtl/ring_counter.v ../tb/ring_counter_tb.v
```

5. Observe the output sequence in the terminal. The ring counter testbench checks the expected sequence, one-hot condition and asynchronous reset operation.

6. View and Analyse the Waveforms Using SimVision

1. After simulation, open the generated waveform file using:

```
simvision dump.vcd
```

2. In SimVision, locate and select the required testbench from the design hierarchy.

3. Select the required input and output signals and add them to the waveform window using the waveform icon.

4. Observe the waveform to verify the operation of the respective design.

7. Verification of Results

For the 4-bit Magnitude Comparator:

• Verify that GT becomes HIGH when A is greater than B.

• Verify that EQ becomes HIGH when A is equal to B.

• Verify that LT becomes HIGH when A is less than B.

• Confirm that only the correct comparison output is HIGH for each test case.

For the 4-bit Ring Counter:

• Verify that the asynchronous reset initializes the output to 0001.

• Observe the sequence 0001 → 0010 → 0100 → 1000 → 0001.

• Confirm that only one output bit remains HIGH during normal operation.

• Verify that reset immediately restores the output to 0001, independent of the clock edge.

• Check the final testbench message: **`RING COUNTER TB : PASSED`**.

Note: *Both supplied testbenches use the filename* `dump.vcd`*. Run each simulation separately and inspect its waveform before running the next simulation, or save the waveform files under different names to avoid overwriting one with the other. Actual execution should be confirmed from the simulator output and waveforms.*